

Diagram illustrating the Tucker decomposition of a 3D tensor T into a core tensor G and three factor matrices A , B , and C .

The tensor T is represented by a purple cube and is associated with the dimensions $\mathbb{R}^{n \times m \times p}$.

The decomposition is shown as a sum of rank-1 components:

$$T = \sum_{r=1}^R a_r \otimes b_r \otimes c_r$$

Each component consists of a blue vertical line segment labeled a_r , an orange horizontal line segment labeled b_r , and a green diagonal line segment labeled c_r .